

Inuka Silva

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EDUCATION

McMaster University (10.3/12)

Candidate for Bachelor of Engineering (B.Eng.), Mechatronics Engineering

Hamilton, ON

Sept. 2025 - Present

- Transferred to pursue studies in Robotics Engineering

University of Toronto

Candidate for Honours Bachelors of Science, Computer Science

Mississauga, ON

Sept. 2024 - April 2025

Chippewa Secondary School

High School Diploma

North Bay, ON

Sept. 2020 - June 2024

- Schulich Nominee, IB, SHAD 2022 Alumnus
- 1 of 8 Canadian FIRST Robotics 2023 Dean's List Finalist

TECHNICAL SKILLS

Languages: Python, Java, R, JavaScript, HTML/CSS, C, C++

Frameworks and libraries: NumPy, pandas, OpenCV, matplotlib, Streamlit, MediaPipe, Next.js, Tailwind CSS

Tools: Onshape, Autodesk Inventor, PrusaSlicer, Git, Photoshop, Illustrator, PyCharm

Soft Skills: Leadership, Teamwork, Presenting

EXPERIENCE

Engineering and Computer Science Summer Intern

May 2025 – Aug. 2025

REDPATH MINING

North Bay, ON

- Designed and fabricated a data logger to help monitor equipment and machine life
- Records IMU data and transmits to a Raspberry Pi using LoRa, Arduino and 3D printed components
- Maintained project documentation to streamline future development

PROJECTS

Robotic End Effector | *Python*

- Designed, fabricated, and tested a 3D printed end effector attachment for a Quanser's QArm
- The end effector was created to pick/drop off unique shaped items autonomously from specified locations
- Warehouse management system was created to log inventory and complete orders

FRC Robot | *Java*

- Assisted with designing, fabrication and assembly of 2022, 2023, and 2024 competition robots
- Integrated autonomous path following and pose estimation with apriltags
- Implemented PID feedback loops and triple modular redundancy for motor control
- Developed code for a swerve drivetrain
- Wrote and implemented code for the 2022, 2023, and 2024 competition robots

3D Printed Watering Can

- Design and fabricated a watering can for a client to reduce wrist usage by using air pressure
- Testing plan was created and executed using KPIs to measure success

Spotify Gesture Controller | *Python*

- Developed a Python application which used hand gestures to control Spotify media functionalities
- Implemented MediaPipe Model Maker to create and train a gesture recognition model
- Used OpenCV library to capture feed from webcam

EXTRA CIRRICULAR

Mentor and Event Volunteer

Jan. 2020 – Present

FIRST Robotics Competition

- Have helped start and mentor FRC teams across Canada
- Volunteered at events across Ontario, helping with field set up, tear down, field reset, and queuing
- Over 500 volunteer and community involvement hours